THE HONG KONG POLYTECHNIC UNIVERSITY
香港理工大學

Department of Building Services Engineering
屋宇設備工程學系

BSE3322 Mini-design Project

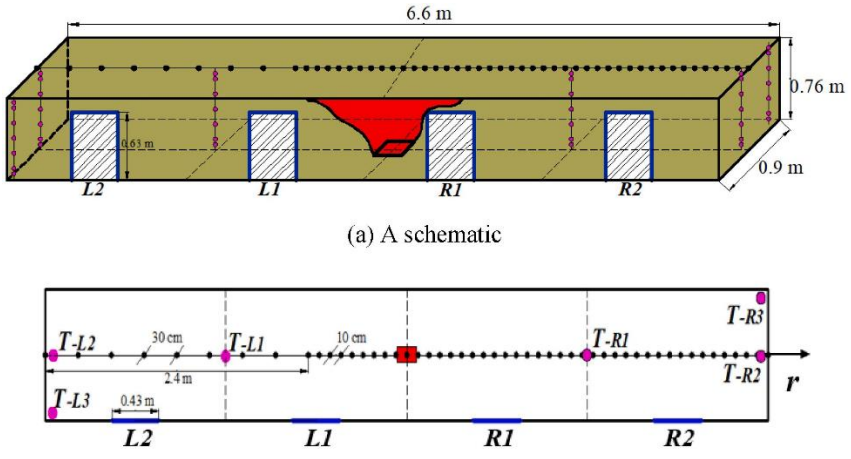
CFD Fire Simulation Study (15 % course total)

Topic B - Natural-ventilated Ceiling Jet Fire Plume Profile in Long Compartment [you can pick topic A OR B]

Problem Description

Ceiling jet fire is a common phenomenon that occurs in compartment fires. It is particularly important in tunnel fire scenarios as the hot gas layer formulated at the ceiling region carries heat to the unburned region. Computational Fluid Dynamics (CFD) technique involving large-eddy-simulation is an effective tool for studying fire plume behaviours. It enables the coupling of non-linear and complex fire phenomena including turbulence, radiation, soot formation and combustion sub-models with consideration of a fire plume's temporal fluctuating/puffing behaviours.

In this project, you will be investigating the ceiling jet behaviour resulting from a fire plume in a reduced-scale long compartmental hall with four openings (i.e. can be closed or opened). The objective was to provide useful data for fire field model validation of a typical ceiling-jet compartment fire scenario in the tunnel. As shown in **Fig. 1,** a three-dimensional long tunnel hallway with 6.6 m long x 0.76 m high x 0.9 m wide is constructed with four openings labelled as L2, L1, R1, R2 (from left to right). The sizes of all openings are 0.43 m wide x 0.63 m high.



(a) A schematic

(b) Top view

Fig. 1. Schematic view and dimensions of the fire test compartment hallway (figure extracted from (Peng et al., 2021)).

Your Tasks

Your tasks are to investigate the heat transfer and flow characteristics of the ceiling jet fire induced by an ascending flame plume. The most important variables you will focus on are the relationship between the inlet and outlet temperatures and velocity, and the overall heat mechanisms including conduction, convection and radiation. In addition, you are required to conduct numerical simulations based on the experiment mentioned with a propane (C_3H_8) burner. The concentration of your nanofluid depends on your Group ID according to **Table 1**.

Table 1. Summary of experiment scenarios, opening settings and HRR.

Group No.	**Test No.**	**Config**	**Opening condition**	**Heat release rate (kW)** (8 cases total based on group no.) (see **bracket** for group no. details)
All	1–8	OP-1	L1, L2, R1, R2	23.6 (**odd**), 47.2 (**even**), 70.8 (**odd**), 106.2 (**even**)
1-4	9-16	OP-2	L1, L2, R2	
5-8	17-24	OP-3	R1, R2	
9-12	25-32	OP-4	L1, R1	
13-17	33-40	OP-5	L2, R2	
18-22	41-48	OP-6	R1	
>22	49-56	OP-7	R2	

In summary, the following are the recommended sub-tasks to accomplish the outcome:

1. Construct the geometry as described in **Fig. 1** (i.e. make sure you set the right boundary conditions, i.e. extended boundaries to capture the doorway profiles);
2. Perform mesh sensitivity analysis on Test no. 1-8, starting from the minimum mesh requirement as suggested by the characteristic length analysis (Yuen et al., 2015);
3. Validate your results against the ceiling temperature profile measured in the literature-reported experiment;
4. Apart from the OP-1 Config, perform simulations on the partially opened OP-2 to OP-7 config based on your group number, and validate your ceiling temperature profile against experimental data;
5. Investigate the effect of various sub-grid-scale (SGS) turbulence models starting from the default model;
6. Examine the effect of at least one of the key modelling constants which govern the flow and heat transfer behaviours.
7. Explain the influences of doorway opening conditions towards the fire plume and the overall ceiling jet profiles.

Design and optimisation

1. Now, consider a propylene flame with a smoke yield of 0.02 and a CO yield of 0.01. Design a smoke extract system using smoke vents and curtains to ensure the smoke layer does not descend below half of the compartment. (If mechanical fans

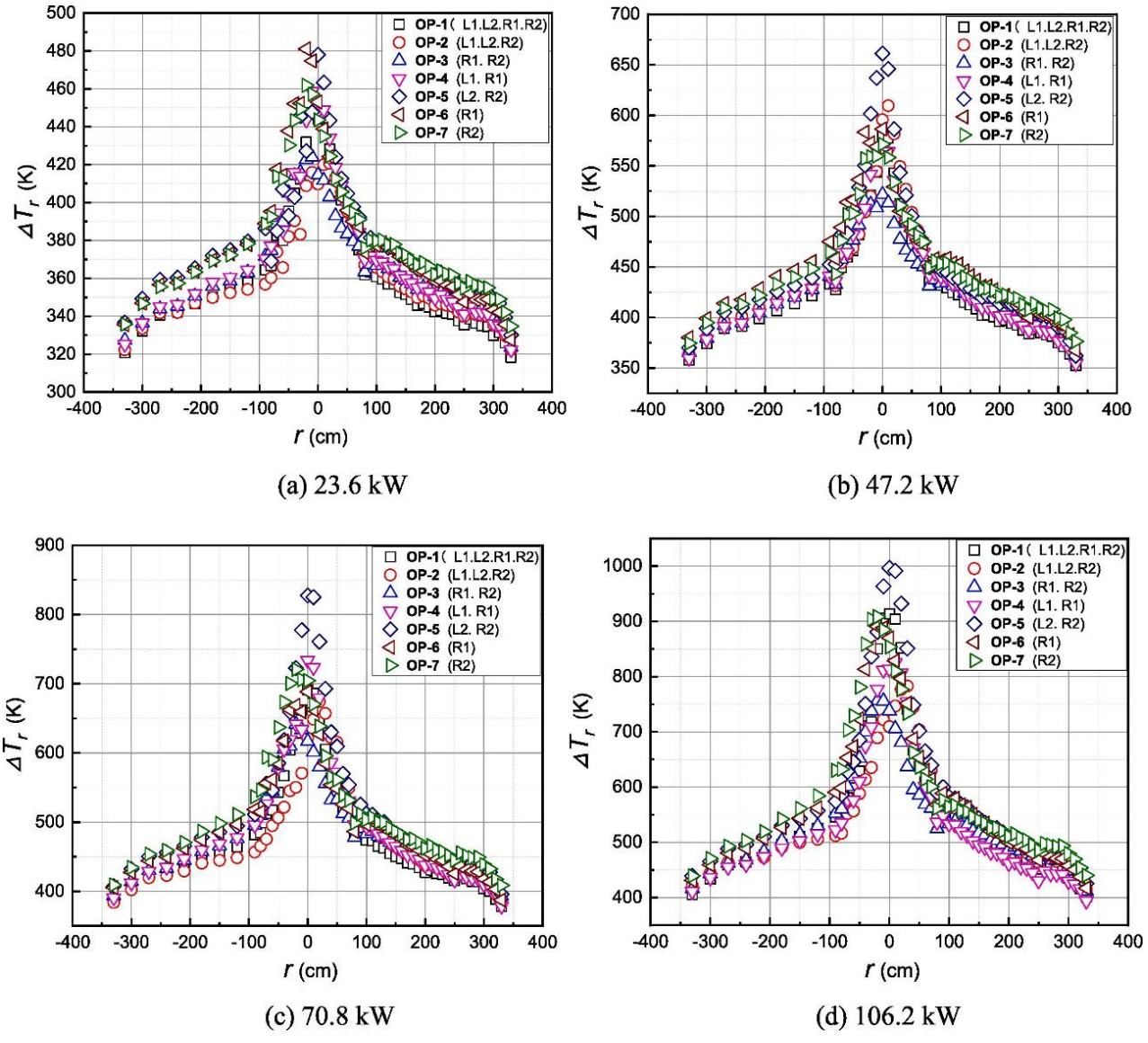
are required, please evaluate the power required)

2. Elaborate on your design options, strategy and stipulated critical case scenarios with more sooty flame conditions (i.e. using a smoke yield of 0.05, 0.1).

3. Comment on the carbon monoxide (CO) generation from the fire and how to mitigate such toxic chemical species.

Data

The tutorial to extract data from the graphs can be found at the following (link).



(a) 23.6 kW (b) 47.2 kW

(c) 70.8 kW (d) 106.2 kW

References

1. PENG, M., CHENG, X., CONG, W., YANG, H., SHAHID, M. U., YUEN, R. & ZHANG, H. 2021. Experimental study on temperature profile in long-narrow compartment fire with multiple lateral openings. *Tunnelling and Underground Space Technology,* 117**,** 104018.
2. YUEN, A. C. Y., YEOH, G. H., TIMCHENKO, V. & BARBER, T. 2015. LES and multi-step chemical reaction in compartment fires. *Numerical Heat Transfer; Part A: Applications,* 68**,** 711-736.